NAINA MITTAL

Practical Test: Prompt Engineering (Beginner Level)

Duration: 45–60 minutes
Tool Allowed: Any LLM (ChatGPT, Gemini, Claude, etc.)

Section A: Prompt Writing (10 Marks)

Question 1

Write a **clear and structured prompt** that asks an AI model to:

> Explain **Machine Learning** to a **class 10 student** using **simple language and real-life examples**.

Expected Skills Tested

- Clear instruction
- Audience specification
- Output clarity

Answer –

Explain the concept of Machine Learning to a Class 10 student who has no prior knowledge of artificial intelligence and machine learning. Use very simple language and avoid technical jargon. Start with a short definition, then explain how it works in simple terms. Include at least three to four real-life examples such as YouTube recommendations, spam email filtering, or face unlock on smartphones. Use bullet points or short paragraphs to keep the explanation clear and easy to understand. Also End with a one-line summary of Machine Learning.

Question 2

Write a prompt that asks an AI model to generate:

- **5 quiz questions**
- Topic: **Artificial Intelligence**
- Difficulty: **Easy**
- Include **answers**

Answer—

Act as a quiz creator for beginners. Generate 5 easy-level quiz questions on the topic Artificial Intelligence. The questions should be simple and suitable for beginners or school students. Use a multiple-choice format with four options (A, B, C, D) for each question. After each question, clearly provide the correct answer. Present the output in a numbered list and keep the language clear and simple.

Section B: Prompt Improvement (10 Marks)

Below is a **poor prompt**:

> "Tell me about AI."

Task:

Rewrite this prompt to make it **more effective and specific**.

Your improved prompt must include:

- Target audience
- Output format
- Examples or explanation

Answer ---

Act as a technology teacher. Explain Artificial Intelligence (AI) to high school students who have little or no prior knowledge of the topic. Use simple and clear language. Start with a short definition of AI, then explain how it works. Provide 2–3 real-life examples such as voice assistants, recommendation systems, or self-driving cars. Present the explanation in bullet points or short paragraphs so it is easy to read and understand

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Section C: Role-based Prompting (10 Marks)

Write a prompt that asks the AI to:

- Act as a **Data Science teacher**
- Explain **Cosine Similarity**
- Include:
  - Formula
  - Simple example
  - Python code snippet

Answer---

Act as a Data Science teacher. Explain the concept of Cosine Similarity in a clear and beginner-friendly way. Start by giving the formula of Cosine Similarity, then explain what each part of the formula means. After that, provide a simple example showing how cosine similarity works between two vectors. Finally, include a short Python code snippet (using libraries like NumPy or scikit-learn) that demonstrates how to calculate cosine similarity between two vectors. Present the explanation in clear steps or bullet points so it is easy for beginners to understand.

Section D: Structured Output Prompt (10 Marks)

Write a prompt that asks the AI to generate a **table** containing:

Algorithm	Use Case	Example

for **5 Machine Learning algorithms**.

Answer—

Act as a Machine Learning instructor. Generate information for 5 Machine Learning algorithms and present the output in a table format with the following columns: Algorithm, Use Case, and Example. Choose commonly used algorithms (such as Linear Regression, Decision Tree, K-Nearest Neighbors, Support Vector Machine, and Random Forest). Provide short and clear descriptions suitable for beginners.

Section E: Prompt Debugging (10 Marks)

An AI produced a **very long and complex answer** when asked a question.

Write a prompt that instructs the AI to:

- Give the answer in **5 bullet points**
- Use **simple language**
- Limit response to **100 words**

Answer---

Answer the following question using simple and easy language. Present the response in exactly 5 bullet points. Keep the explanation clear and concise, and limit the total response to a maximum of 100 words.

Bonus Question (Optional – 5 Marks)

Write a prompt that asks an AI to generate a **Python program that performs sentiment analysis on text**.

Answer---

Act as a Python and Data Science expert. Generate a Python program that performs sentiment analysis on text. The program should take a text input and classify it as Positive, Negative, or Neutral. Use a simple library such as TextBlob . Include clear comments in the code explaining each step, and provide a short example showing how to run the program with sample input text

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